Hildebrand: exact finite box counting

Result. Lean proves the finite counting core for genuine uniform sampling without replacement. The source box probability is at most the product of coordinate bounds p·(U−L)+1 divided by (p−1).descFactorial d, for every prime p, d≤p−1 and nonzero Fourier index j. The actual probability law, cardinality and event are bound in the main theorem. [pcc:unverified]

Actual sampler	Uniform injective d-tuples; d ≤ p−1 Full conditional supports: cardinality (p−1).descFactorial d
Exact finite bound	Nonzero j: coordinate count ≤ p·(U−L)+1 Probability ≤ product of counts / falling factorial
Negative control	Index j=0, d=1, L=U=0: probability 1 For p≥3, this exceeds deleted-index bound 1/(p−1)

Sampler and normalization. Actual injective d-tuples from a finite population X have cardinality (card X).descFactorial d. Lean derives actual uniform point masses, the exact event/cardinality ratio and a coordinate-product upper bound by injection into the full coordinate product. No independence premise is assumed. If d exceeds the population, Lean proves that no PMF on the sample space exists. [pcc:unverified]

Full source space and geometry. An explicit equivalence identifies injective tuples of nonzero prime-field elements with all injective support maps with fixed first point a_0. The mapped sampler equals the actual uniform law on this full conditional space. Its source expression $(a_0-a_i)\cdot j$ is carried through the equivalence. For j nonzero, transformed canonical centered residues form an injective integer image. Actual finite integer-interval counting gives width+1; the lattice correction is retained. [pcc:unverified]

Deleted-index negative control. For every prime p≥3, j=0 makes each transformed centered coordinate zero. A zero-width coordinate box has p−1 choices, exceeding its width-plus-one bound1. For d=1, its actual full-source event probability equals1 and is strictly greater than1/(p−1). Lean directly negates the probability inequality with the nonzero-index hypothesis deleted. This does not refute the qualified lemma. [pcc:unverified]

Source review and remaining work. Hildebrand1994, proof p197 supplies the exact product(width+1) numerator and falling-factorial denominator; full13-page original read and inspection credited. Author preprint math/0401189v1, selected section4.2/pages13–16/reference14p36, restricts the later asymptotic lemma to k≥3; visual15–16 only. The 1.1 asymptotic still needs width-growth and denominator estimates; the stronger expected random-support mixing exponent remains open. [pcc:unverified]

Checks. Whole-project gate: 49 declarations, zero failures, zero trusted native exceptions. All 131 build/processing/provenance files retained losslessly in 46 objects, plus five gate files and recursive source/config pins. Lean4.35.0/Mathlib56a00585…, standard foundations/default limits. Signed standing unverified. KB/Conductor unavailable; bounded fallback recorded before code. No holes/custom axioms/native evaluation/raised limits/global flags/waivers/native experiment/paid compute/upload. [pcc:unverified]